

Training to Get a Drone License, Virtually

Ming-Ru Xie
Department of Computer Science,
National Yang Ming Chiao Tung
University, Taiwan
mingruxie.cs05@nycu.edu.tw

Siang-Jhih Sie
Department of Computer Science,
National Yang Ming Chiao Tung
University, Taiwan
souleye1983@gmail.com

Shing-Yun Jung
Department of Computer Science,
National Yang Ming Chiao Tung
University, Taiwan
syjung@cs.nctu.edu.tw

Bo-Wei Dong
Department of Computer Science,
National Yang Ming Chiao Tung
University, Taiwan
bwdong.cs07@nycu.edu.tw

Pei-Hua Tsai
Department of Computer Science,
National Yang Ming Chiao Tung
University, Taiwan
phtsai.cs07@nycu.edu.tw

Neng-Hao Yu
Department of Design, National
Taiwan University of Science and
Technology, Taiwan
jonesfish@gmail.com

Kuan-Wen Chen
Department of Computer Science,
National Yang Ming Chiao Tung
University, Taiwan
kuanwen@cs.nctu.edu.tw

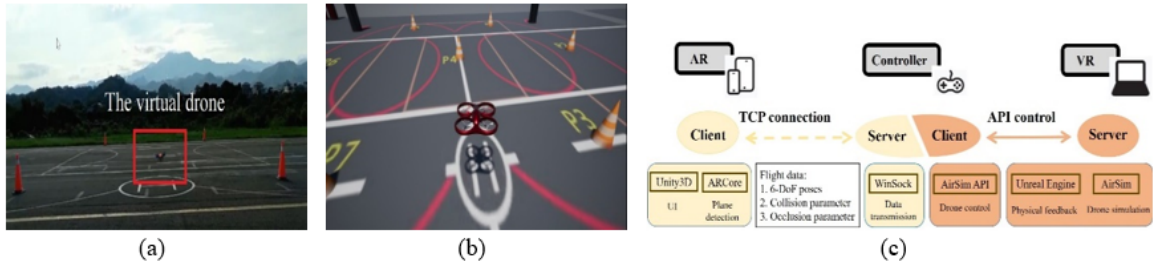


Figure 1: An example of (a) MR view and (b) its corresponding VR environment, and (c) DFT MR system architecture.

ABSTRACT

We propose a drone flight training system (DFT MR) with a novel mixed reality (MR) technique to lower the risk of drone crashes for drone flight beginners and enhance their actual feelings of drone control. Recently, virtual reality (VR) or augmented reality (AR) has been widely employed in training systems. However, the current drone training systems in AR usually lack physical feedback from wind or air resistance. In addition, the VR training environment and the real world were considered to be inequivalent. The proposed MR system integrates the advantages of both VR and AR by transmitting the high-fidelity simulated physical control feedback, such as obstacle collision and wind influence, from VR to the MR device. The system not only benefits those who have VR sickness but also provides novice drone operators virtual crash-free drones for training to get their drone licenses.

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1 INTRODUCTION

Recently, a drone license or certificate has become compulsory internationally for outdoor drone operators. Drone flight training tends to be indispensable in terms of drone license exam preparation. However, the risk of drone crashes may still be high for novice drone operators who have not passed the license exam yet, especially in their flight training process. Hence, constructing a safe and crash-free training environment for drone flight beginners is an issue to be addressed.

VR or AR environments have been used for training in recent studies. However, the VR training environment and the real world may not be equivalent. The time of completing training tasks was indicated significantly different between VR and real-world environments [Read and Saleem, 2017]. In addition, VR may cause virtual reality sickness [CHANG et al., 2020]. Unlike those of VR, as far as we know, there are only a few applications of AR in drone

Table 1: Evaluation results for expert versus novice drone operator.

	Expert*			Novice user**		
Effectiveness	Strongly agree	Agree	Neutral	Strongly agree	Agree	Neutral
	75%	25%	0%	57.1%	35.7%	7.1%
Realism	Strongly agree	Agree	Neutral	Strongly agree	Agree	Neutral
	62.5%	37.5%	0%	53.6%	35.7%	10.7%

* Eight of the participants are experts. ** Fourteen of the participants are novice users.

flight training with no real-world physical feedback. Some drone training systems have recently applied MR technology [KIM et al., 2018; GO et al., 2020] for real and virtual interaction. However, they use a real drone to conduct drone training in an MR environment, blending the virtual and real world. Real drones require open safe spaces and enough battery time, and most importantly, the risk of drone crashes still exists when any operations go wrong.

This paper proposes a novel MR technique based on VR and integrates the advantages of AR to construct a virtual drone flight training system (DFT MR) (Figure 1) for the Remote Pilots License Practical Test preparation in Taiwan. The flight environment created by this system allows novice drone operators to practice operating a virtual drone securely until they become proficient operators. The system includes an MR view with a virtual drone in the real environment (Figure 1(a)). We create a simple virtual scene in the VR environment according to the real environment (Figure 1(b)) to reduce the computational cost of providing physical control feedback in the MR device. Then, the system simulates the physical feedback through VR and transmits the simulation results to the MR device. As our VR environment focuses on providing physical feedback, it requires only inexpensive computing devices to run. Both statuses of drones in MR and VR are synchronized.

2 SYSTEM ARCHITECTURE

Figure 1(c) shows the system architecture of the proposed DFT MR. It contains two parts: MR site and VR site. The VR site is built using Unreal Engine (providing physical feedback) and AirSim (providing drone control). A monitoring program is plugged in at the VR site to record the 6-DoF (degrees of freedom) poses of the VR drone, collisions, and the occlusion status, along with the other parameters. The monitoring program will then transmit these data to the MR site in time via a TCP connection. The MR site is built using ARCore and Unity3D, and a TCP server is embedded. ARCore and Unity project a corresponding virtual drone in a real scene according to the drone flight status of the VR site synchronously. Furthermore, there are two additional boolean parameters for collision and occlusion. They are calculated based on the position of the drone and the obstacle in AirSim. The collision parameter indicates whether the drone collides with an obstacle. When it happens, the drone in MR will stop moving. The occlusion parameter determines whether the drone is occluded by obstacles. When the occlusion parameter is true, the drone in MR will disappear behind the obstacle. A demo video of DFT MR can be found on the following link: https://youtu.be/7Wk_rHpMTbs.

3 EXPERIMENT

To evaluate our training system, we designed a test environment based on the real Remote Pilots License Practical Test of the Taiwan Civil Aviation Administration (CAA). We first measured and created a 1:1 virtual environment and set up obstacles in the VR site. Thereafter, we invited 22 participants to join the drone flight test at the formal test site. We set the Unreal Engine's wind force according to the weather situation on the day of the experiment. After the experiment, we invited the participants to fill out the training system's questionnaire and conducted interviews.

Table1 shows the evaluation results. Here, effectiveness means how the subject agrees that this system could improve the training effectiveness. Realism means how the subject agrees that the drone control in MR was realistic. We divided participants into two groups: experts and novice users according to their experiences of license exams. Experts mean the participants have taken the drone license exams, and novice users mean the participants have not taken drone license exams. The results show that 75% of the experts strongly agreed that the system was beneficial for the license exam. Another 62.5% of experts strongly agreed that the virtual drone had a high degree of physical feedback to make them feel very real. Three of the experts even suggested in the interview that we cooperate with CAA to use our system for training. In summary, we cared the evaluations of effectiveness from the expert group more. They gave us a lot of positive feedback on our system. Besides, most of the experts indicated that the simulation of the wind influence plays a key role in training and making the system more realistic.

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